

# Part 2 Study Guide

## Topic 7: Geometric Sequences

### GEOMETRIC SEQUENCE FORMULA:

- 1) Given the sequence  $\{-2, 6, -18, \dots\}$ , write a rule, then find  $a_8$ .

$$a_n = -2(-3)^{n-1}$$

$$a_8 = -2(-3)^7 = \boxed{4374}$$

- 2) Given the sequence  $\{6144, 3072, 1536, \dots\}$ , write a rule, then find  $a_{15}$ .

$$a_n = 6144\left(\frac{1}{2}\right)^{n-1}$$

$$a_{15} = 6144\left(\frac{1}{2}\right)^{14} = \boxed{.375}$$

## Topic 4: Scientific Notation

Evaluate. Give each final answer in scientific notation.

3)  $(8.6 \times 10^{-4}) + (6.7 \times 10^{-5})$   
 $8.6 \times 10^{-4} + .67 \times 10^{-4}$

$$\boxed{9.27 \times 10^{-4}}$$

4)  $(3.5 \times 10^3) - (9 \times 10^1)$   
 $3.5 \times 10^3 - .09 \times 10^3$

$$\boxed{3.41 \times 10^3}$$

5)  $(5.6 \times 10^4) \times (4.5 \times 10^6)$   
 $25.2 \times 10^{10}$

$$\boxed{2.52 \times 10^{11}}$$

6)  $(7.6 \times 10^{-7}) \times (8.9 \times 10^{-2})$   
 $67.64 \times 10^{-9}$

$$\boxed{6.764 \times 10^{-8}}$$

7)  $(3.9 \times 10^{-12}) \div (4 \times 10^4)$   
 $.975 \times 10^{-16}$

$$\boxed{9.75 \times 10^{-17}}$$

8)  $\frac{(1.75 \times 10^5)}{(5 \times 10^{-3})}$   
 $.35 \times 10^8$

$$\boxed{3.5 \times 10^7}$$

- 9) The continent of North America has an area of approximately  $9.4 \times 10^6$  square miles. The area of Asia is approximately  $1.74 \times 10^7$  square miles. Approximately how many square miles larger is Asia than North America?

$$1.74 \times 10^7 - .94 \times 10^7$$

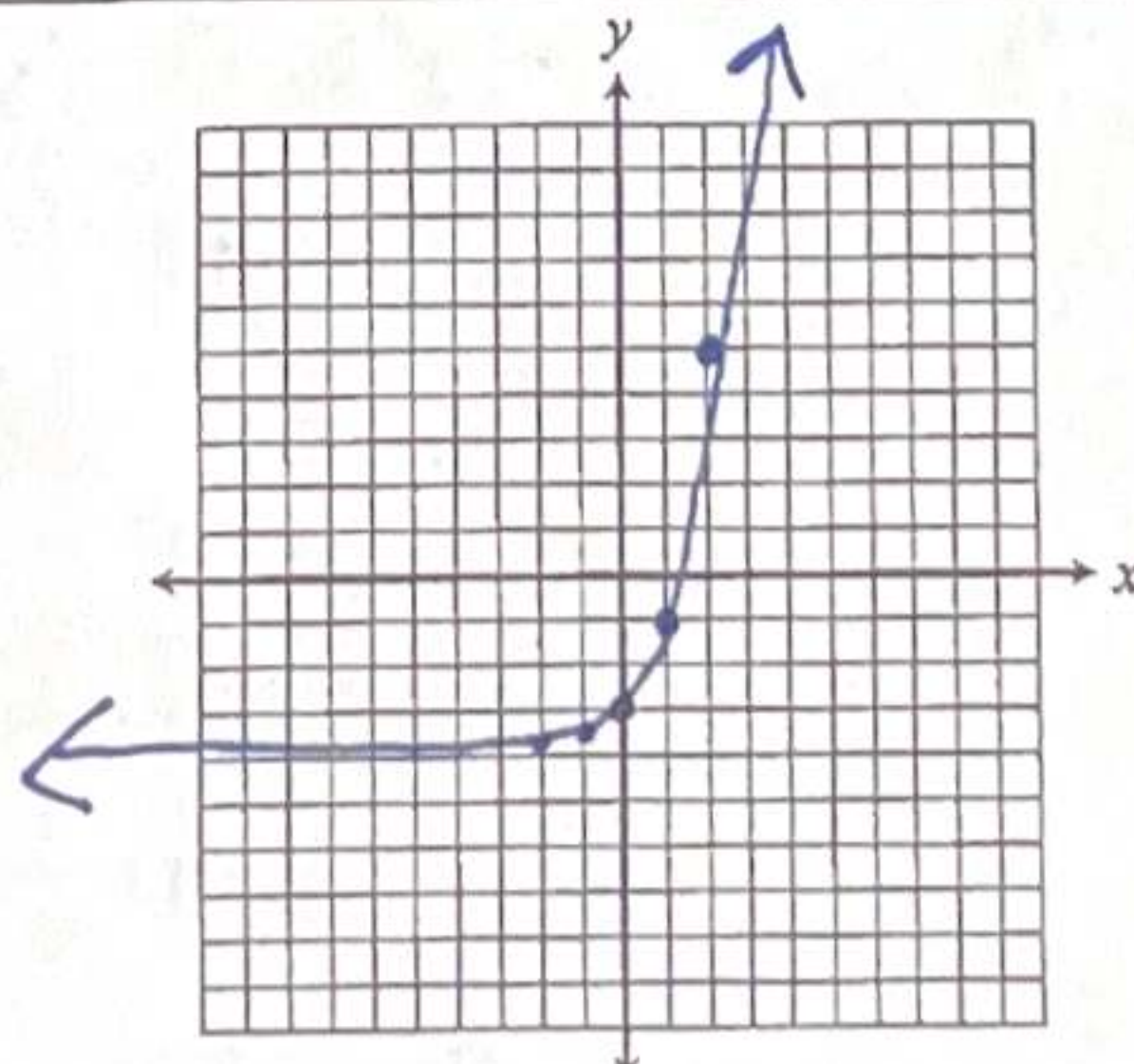
$$.8 \times 10^7 = \boxed{8 \times 10^6}$$

## Topic 5: Graphing Exponential Functions

Graph each function using a table of values, then identify its key characteristics.

10)  $y = 3^x - 4$

| x  | y    |
|----|------|
| -2 | -3.8 |
| -1 | -3.6 |
| 0  | -3   |
| 1  | -1   |
| 2  | 5    |



Growth / Decay

Domain:  $\mathbb{R}$

Range:  $y > -4$

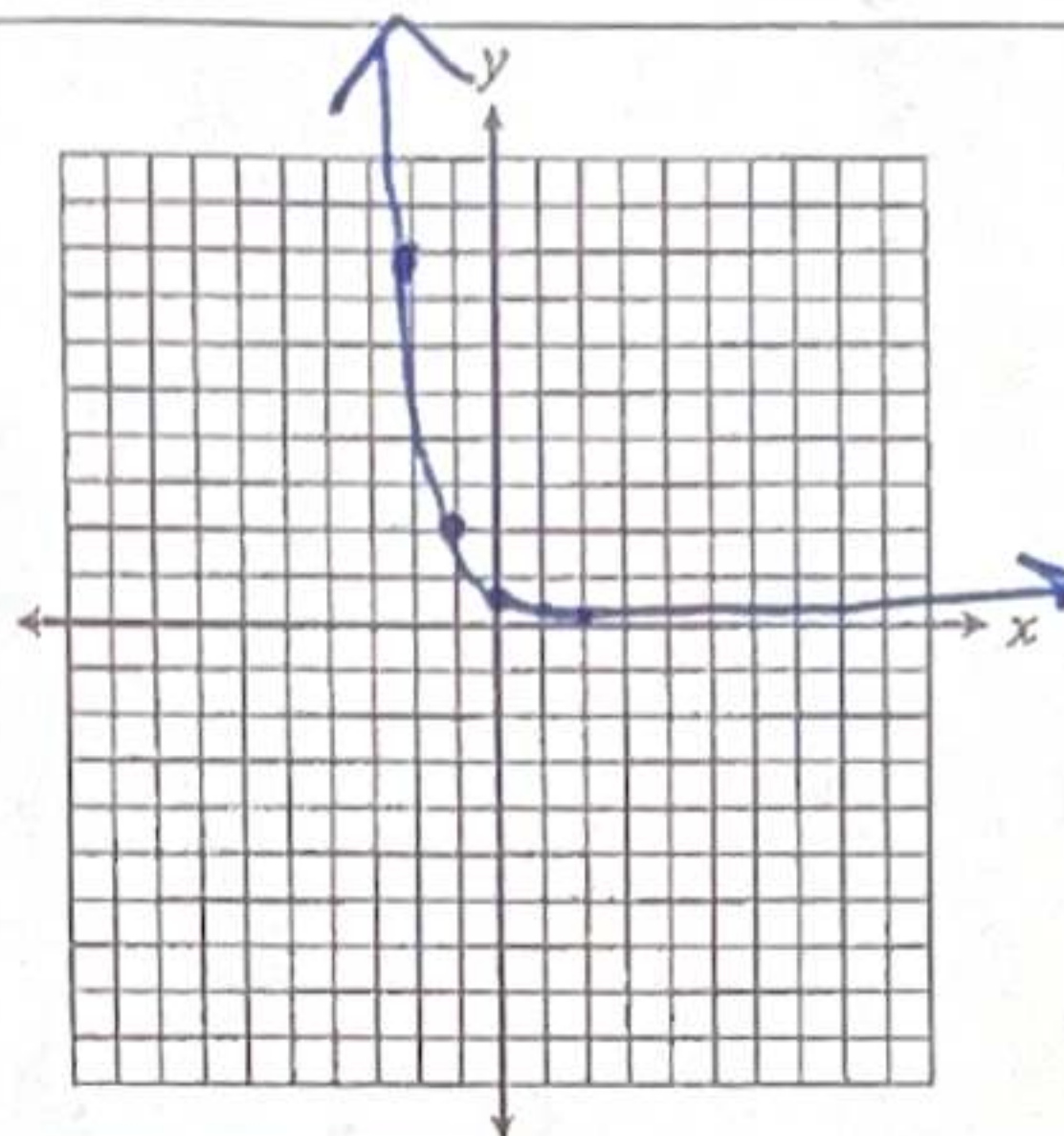
y-intercept:  $(0, -3)$

Asymptote:  $y = -4$



$$11) y = \frac{1}{2} \cdot \left(\frac{1}{4}\right)^x$$

| x  | y      |
|----|--------|
| -2 | 8      |
| -1 | 2      |
| 0  | .5     |
| 1  | .125   |
| 2  | .03125 |



Growth / Decay

Domain:  $\mathbb{R}$

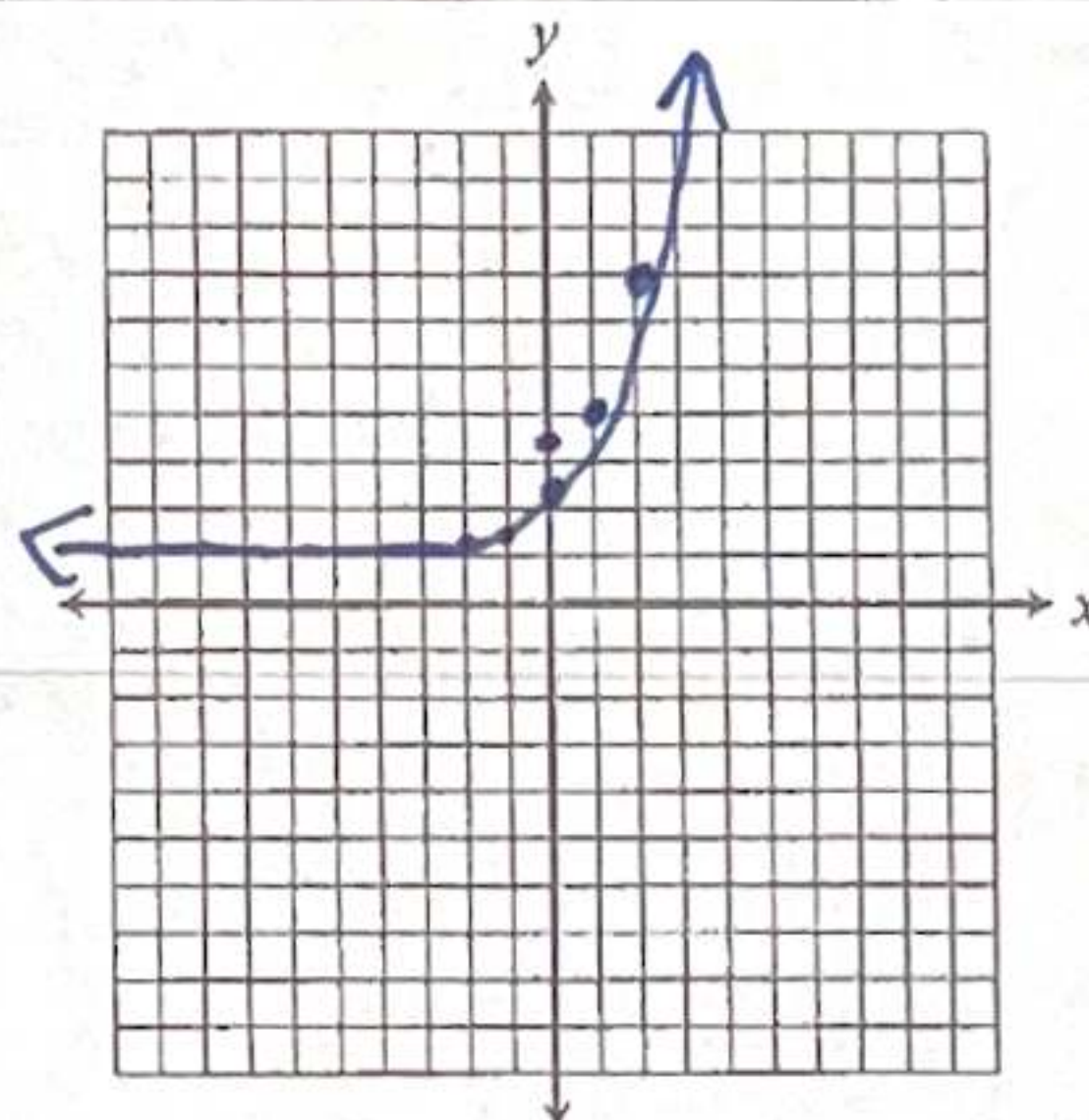
Range:  $y > 0$

y-intercept:  $(0, .5)$

Asymptote:  $y = 0$

$$12) y = \frac{3}{2} \cdot 2^x + 1$$

| x  | y     |
|----|-------|
| -2 | 1.375 |
| -1 | 1.75  |
| 0  | 2.5   |
| 1  | 4     |
| 2  | 7     |



Growth / Decay

Domain:  $\mathbb{R}$

Range:  $y > 1$

y-intercept:  $(0, 2.5)$

Asymptote:  $y = 1$

#### Topic 6: Exponential Growth & Decay Applications

##### EXPONENTIAL GROWTH FUNCTION

$$y = a(1+r)^t$$

##### EXPONENTIAL DECAY FUNCTION:

$$y = a(1-r)^t$$

- 13) A population of a city is 422,000 and increases by 12% each year. Use an exponential function to find the population of the city after 8 years.

$$y = 422000(1+.12)^8$$

$$\boxed{y = 1,044,856 \text{ people}}$$

- 14) A car bought for \$13,000 depreciates at 15% per year. Use an exponential function to find the value of the car after 5 years.

$$y = 13,000(1-.15)^5$$

$$\boxed{y = \$5768.17}$$

- 15) Scott purchased a painting in 2006 for \$1,250. Since then, its value has increased by 6% each year. Use an exponential function to find the value of the painting in 2017.

$$y = 1250(1+.06)^{11}$$

$$\boxed{y = \$2372.87}$$